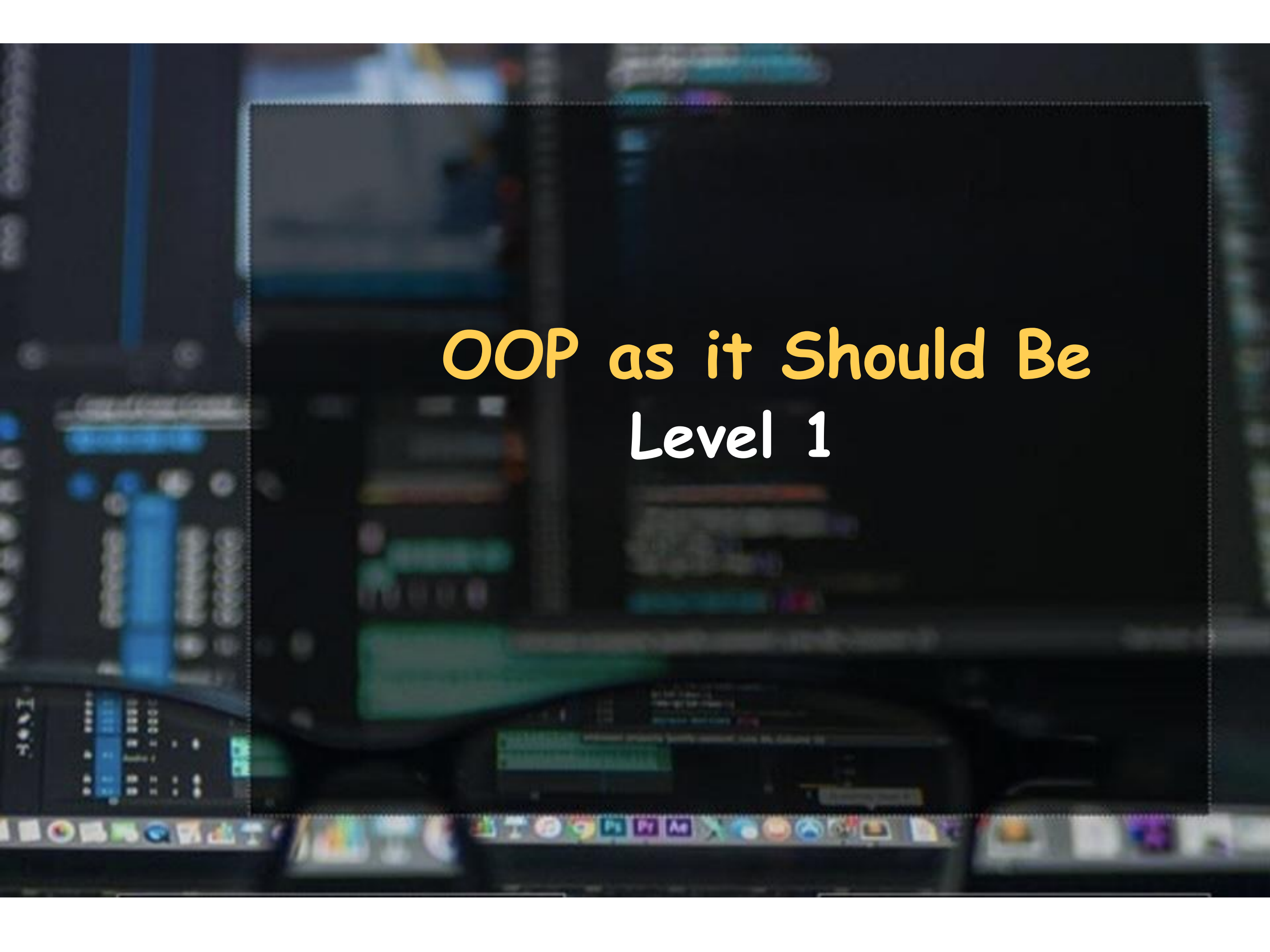




OOP as it Should Be Level 1



OOP as it Should Be

Level 1

Read Only Property

Definition

This example explains the concept of a Read-Only Property in C++, where a class attribute can be accessed but cannot be modified directly.

Explanation

- ◆ A read-only property has a getter only
- ◆ No setter function is provided for that property
- ◆ This prevents accidental or unauthorized modification of the value
- ◆ It is a common practice for IDs and system-controlled values

Components of clsPerson class

① Private Data Members

- `_ID` : initialized with value 10
- `_FirstName` : stores first name
- `_LastName` : stores last name

② Read-Only Property

- `ID()`
- Returns `_ID`
- No `setID()` function exists
- `_ID` cannot be changed from outside the class

③ Setter Functions

- `setFirstName(string FirstName)`
- `setLastName(string LastName)`
- Used to assign values to private members

Components of clsPerson class

4 Getter Functions

-  FirstName()

-  LastName()

-  Used to retrieve stored values safely

5 FullName Function

-  FullName()

-  Combines first name and last name into one string

Example

- ◆ ID is accessed
- ◆ ID cannot be modified

```
clsPerson Person1;  
Person1.setFirstName("Sara");  
Person1.setLastName("Alterkawi");  
cout << Person1.ID();
```

Steps in main()

- 1 Create object Person1
- 2 Set first name
- 3 Set last name
- 4 Read ID
- 5 Print first name
- 6 Print last name
- 7 Print full name

Expected Output

ID: 10

First Name: Sara

Last Name: Alterkawi

Full Name: Sara Alterkawi

Note

Even though `_ID` is initialized, it cannot be changed because there is no setter function.



Comparison



Read-Only Property

- ✓ More secure
- ✓ Prevents accidental changes
- ✓ Ideal for IDs



Read/Write Property

- ✗ Can be modified
- ✗ Less control

Warning

Do not make sensitive values public just to access them—use a read-only getter instead.

Summary

- ✓ Read-only property = *Getter* without *Setter*
- ✓ Protects important data
- ✓ Enforces encapsulation
- ✓ Common in professional C++ design

Thank you 😊